



SpotLite: A System for Personalized, Aspect-Based Restaurant Recommendation Near Tourist Destinations

Group 31: Ting-Long Wei, Shih-Hui Huang, Sheng-Kuo Lin,
Suihan Gao, Jingyi Xia, Yu-Chen Lu

Outline

Motivation

Research Questions

Proposed Methods

Findings

Contribution

Motivation



Why SpotLite?

Commercial platforms like Google Maps suffer from three major limitations:

- **Information Overload:** Users must manually read through long, unstructured reviews to find what they care about.
- **Lack of aspect-level insights:** Users currently need to manually scan through paragraphs of text to find feedback related to specific aspects such as food, service, or price.
- **Unclear sentiment:** While Google Maps highlights frequently mentioned keywords, these keywords do not indicate whether the sentiment is positive or negative.

Research Questions Answered



Research Questions Answered in the Project

1. How can we convert unstructured review text into structured, aspect-level insights
2. How can we assign reliable sentiment evaluations to user opinions
3. How can we generate grounded, interpretable, aspect-aware keywords and summaries

Proposed Methods



01

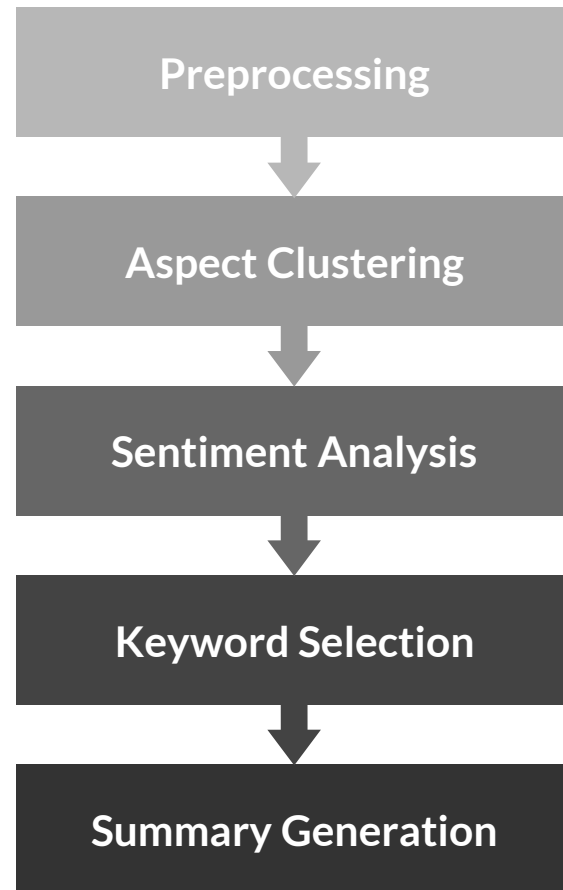
Data Collection & Filtering

- Scrape Google Maps restaurant reviews in the LA area using Selenium
- Filter restaurants with ≥ 50 reviews, and retain only reviews from the past year
- Construct a high-quality corpus of $\sim 1,000$ restaurants



02

NLP Pipeline





Preprocessing

```
{  
  "review_id": "Ci9DQUlRQUVhZm9udHJlRjlvT250MlNHcFFkR3hEUVRSd1ZWZDRSVEJwZUkxOVRiRWhefction."  
  "sentence": "T",  
  "aspect_ratings": {  
    "food": 5.0,  
    "service": 5.0,  
    "atmosphere": 5.0  
  },  
  "reviewer": "Mike Coplin",  
  "stars": "5 stars",  
  "date": "a day ago"  
},
```

1. Raw Review Scraping → JSON Format

Each scraped Google Maps review is stored as JSON object (with reviewer_id and relative information). The `text` field is the primary NLP input.

1. Extracting Google's New Review Template

Detect and strip template lines (Inline review scoring about different aspects) and store them in a separate meta json file (per store).

1. Cleaned Text + Template Data = Dual Inputs

Cleaned text → used for NLP for sentence-level sentiment, aspect tagging, keyword extraction.

Metadata → Used as support data for rating histograms, price, dishes recommendations etc.



Aspect Clustering

Automatically group review sentences into meaningful aspects and extract top key phrases per aspect. Cluster into 5 hand-defined aspects: Taste, Service, Environment, Waiting Time, Price.

1. Sentence Transformer Embeddings

Convert each review sentence into vector embeddings using MiniLM (all-MiniLM-L6-v2)

1. Prototype-based Aspect Assignment

Compare each sentence embedding to 5 aspect prototype embeddings + small keywords sets, and assign the sentence to 0-N aspects based on similarity thresholds.

1. Aspect-aware Phrase Mining

For each aspect, mining n-gram phrases from tagged sentences and score it by review-level support and frequency. Prefer multi-word phrases to provide more informative results.



Sentiment Analysis

- **Model Used:** distilbert-base-uncased-finetuned-sst-2-english (Hugging Face)
- **What We Do:**
 - Run sentence-level sentiment on cleaned review text
 - Convert the model scores into Positive / Neutral / Negative labels using thresholds
 - Aggregate sentence sentiment to per-aspect statistics and separate positive vs negative keyword sets for each aspect.
- **Why This Model?**
 - Lightweight & Fast
 - Lower Computation Costs
 - Easy Integration



Keywords Selection

Extract the most important aspect-specific keywords from reviews.

1. TF-IDF Vectorization

Convert reviews to term-frequency vectors → score each keyword (biagram) by importance.

1. Top Keywords

Select top N keywords by TF-IDF score.

1. Representative Example

For each keyword, pick a review snippet where it appears most strongly.

1. Clustering Keywords

Group similar keywords by co-occurrence in reviews using a PMI-like similarity → agglomerative clustering.



Summary Generation

Automatically generate concise review summaries using extracted aspect keywords.

1. **Model:** Fine-tune T5 (small/base) model using Seq2Seq training
2. **Dataset:** Yelp Open Dataset
 - Convert reviews into third-person summaries (“I loved it” → “Customers loved it”).
 - Build (keywords → summary) pairs
1. **Training Workflow:**
 - Tokenize keyword strings (input) and third-person summaries (target)
 - Shuffle & split data (90% training / 10% validation)
 - Train Seq2Seq model (batch size, epochs, max length defined)
 - Evaluate and save the fine-tuned model + tokenizer

Evaluation

Evaluation Method

We use ChatGPT as an external evaluator.



Holbox_reviews.json
File



model_output.json
File

You are an expert reviewer for an academic NLP project.

I will give you:

- 1) A restaurant review dataset (raw text from reviews.json).
- 2) The output generated by my own non-LLM NLP pipeline, including:
 - Summary for the restaurant
 - Keywords for 5 aspects (Taste, Service, Environment, Waiting Time, Price)
 - Dish highlights extracted from reviews

Your task is to evaluate the quality of my pipeline output.

PART 1: INPUT DATA

[RAW REVIEWS — Holbox_reviews.json]

[PIPELINE OUTPUT — model_output.json]

PART 2: EVALUATION TASK

Please evaluate my output based on the following 5 dimensions:

1. Aspect Accuracy (0–10)
 - Are the aspect categories correctly summarized based on the reviews?
 - Did the pipeline assign information to the correct aspects?
2. Keyword Quality (0–10)
 - Are the extracted keywords relevant and meaningful?
 - Do they reflect real customer concerns, not just generic or trivial words?
3. Coverage & Completeness (0–10)
 - Does the summary capture the main ideas and recurring themes from the reviews?
 - Are any important aspects or insights missing?
4. Noise & Irrelevance (0–10)
 - How clean is the output? - Are there useless, generic, or incorrect keywords?
 - Is there topic pollution across aspects?
5. Dish Extraction Quality (0–10)
 - Are the extracted dish names real dishes or just generic food words?
 - Do they represent what customers frequently mentioned?



Evaluation Result

Dimension	Score	Reason
Aspect Accuracy	8.0	Aspect assignments align well with customer narratives and are strongly improved by TF-IDF keywords, with minor loss of nuance.
Keyword Quality	8.5	Keywords are specific, semantically meaningful, and closely match real review language, with minimal noise.
Coverage & Completeness	7.5	Most core themes are covered, but prestige, emotional intensity, and cleanliness details are underrepresented.
Noise & Irrelevance	7.0	Noise is relatively low, though some mechanical phrasing and sentiment collisions remain.
Summary Quality	6.5	Summary is factually correct but lacks richness, narrative flow, and several high-frequency themes like Michelin recognition and food hall context.

Final Output

restaurants_query API

```
{
  "location_url": "",
  "geolocation": {
    "lat": 34.0224,
    "lng": -118.2851
  },
  "search_radius_meters": 100,
  "preferences": {
    "must_have": [""],
    "nice_to_have": [""],
    "exclude_keywords": [""],
  }
}
```

input

```
{
  "id": 1,
  "name": "Cafe Dulce (USC Village)",
  "matchScore": 92,
  "rating": 4.6,
  "reviews": 495,
  "distance": "N/A",
  "priceLevel": "$$",
  "isOpen": true,
  "image": "https://lh3.googleusercontent.com/gps-cs-s/AG0i5wz7jwYrEYmPlI2dmKaat-X6a9kn1CSU59p8CZMPB_gPgeI04",
  "ai_summary": "A bustling USC Village cafe known for inventive items like the Matcha Latte and Breakfast Burri",
  "address": "3806 McClintock Ave Ste 142B, Los Angeles, CA 90007, USA",
  "phone": "(213) 536-5609",
  "website": "N/A",
  "hours": "Open Now • Closes 8:00 PM",
  "signatures": [
    { "name": "Matcha Latte", "count": 19 },
    { "name": "Breakfast Burrito", "count": 10 },
    { "name": "Pastrami Sandwich", "count": 5 }
  ],
  "aspects": {
    "taste": {
      "rating": 4.6,
      "summary": "Inventive & Delicious Donuts/Coffee",
      "keywords": [
        { "text": "Matcha Latte", "count": 9, "type": "pos" },
        { "text": "Spicy Chicken Sandwich", "count": 2, "type": "pos" },
        { "text": "Kale Peanut Salad", "count": 2, "type": "pos" },
        { "text": "Too Sweet", "count": 3, "type": "neg" },
        { "text": "Nothing Special", "count": 2, "type": "neg" },
        { "text": "Missing Meat", "count": 1, "type": "neg" }
      ]
    }
  }
}
```

output

Contribution of each member

Contribution

- Ting-Long Wei: Data Collection, Full NLP Pipeline
- Shih-Hui Huang: Full NLP Pipeline
- Sheng-Kuo Lin: Full NLP Pipeline
- Suihan Gao: Full NLP Pipeline
- Jingyi Xia: Full NLP Pipeline, Evaluation
- Yu-Chen Lu: Full NLP Pipeline

Thank You



Questions

Findings



Improvements in Aspect Clustering & Keyword Extraction

1. Aspect Clustering

- Started with manually selected aspect keywords
- Improved using embedding-based prototype clustering for more comprehensive coverage and less human bias

2. Keyword Extraction

- Initially relied on part-of-speech patterns (e.g., Noun + Noun, Adjective + Noun), leading to repetitive terms (e.g., price, prices, pricing)
- Improved by applying TF-IDF with domain-specific stopwords and co-occurrence clustering, selecting top keywords and grouping them by similarity



Adapted Aspect Seeds

1. Manually define initial seed dictionary for each aspect
2. Extract candidate aspect phrases from reviews
3. Compute semantic similarity between candidate phrases and aspect prototype embeddings
4. Automatically expand seed dictionary when similarity $\geq$ threshold